

Deploying Large Language Models on Kubernetes

Kubernetes provides a robust platform for deploying and managing large language models, enabling the development of reliable, scalable, highly available, portable and secure applications. There are many benefits of deploying large language models on Kubernetes, and a few challenges too.



Large language models (LLMs) are generative AI models that use deep learning techniques for a wide range of applications like chatbots, content generation, and language translation. LLMs perform natural language processing (NLP) tasks like language translation, text classification, sentiment analysis, text generation, and question-answering.

LLMs are trained with a huge number of datasets from various sources. Some of the most successful LLMs have hundreds of billions of parameters, and could fundamentally transform how we handle and interact with data, including master data.

Some well-known language models are: Gemini from Google, OpenAI's GPT-4, Claude 3 from Anthropic, Bloom developed by a large team of researchers and engineers from

academia and the private sector, with support from the French public supercomputer Jean Zay; and XLNet from Google with a whopping 175 billion parameters.

What is Kubernetes?

Kubernetes is an open source platform designed to automate the deployment, scaling, and management of containerised applications. It provides tools for deploying applications, scaling them as needed, managing changes to existing deployments, and monitoring the health of the system. Its key features are:

Container orchestration: Manages the lifecycle of containers, including deployment, scaling, and updates.

Automated rollouts and rollbacks: Ensures smooth updates and quick recovery from failures.